Optimization Strategies for Local Package Delivery Operations

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1 Introduction

This project is about making local package delivery more cost-effective by reducing the overall travel distance of a small number of vehicles, but still meeting all delivery requirements. The packages are in different locations, weights and have various priorities that need to be considered when assigning them to vehicles which must stay within their load limits.

The job is shaped as a Capacitated Vehicle Routing Problem (CVRP), which people know to be NP-hard. For solving it in an effective way, we use two metaheuristics: Simulated Annealing (SA) and Genetic Algorithm (GA). Both of these are designed to give high-quality solutions within decent time.

The report describes the issue arrangement, solution creation, algorithm information, limitation management and impact of adjusting main parameters.

2 Problem Formulation

We model the delivery task as a CVRP with one depot.

2.1 Inputs and Representation

- **Depot:** Located at (0, 0).
- **Packages:** Defined by ID, location (x, y), weight (kg), and priority (1–5).
- **Vehicles:** N identical vehicles, each with a fixed weight capacity.
- **Distance:** Calculated using Euclidean metric.

2.2 Objective Function

Minimize total distance traveled by all vehicles:

$$D = \sum_{v=1}^{N} \text{distance}(R_v)$$

where R_v is the route for vehicle v, starting and ending at the depot.

2.3 Constraints

1. Vehicle load must not exceed capacity.
2. Each package must be delivered by exactly one vehicle (unless unassignable).
3. Routes start and end at the depot.

2.4 Priority Handling

Priority is a soft factor, used optionally for tie-breaking or initialization. The core optimization minimizes distance.

2.5 Solution Representation

A solution is a list of vehicle routes, each containing a sequence of package IDs, e.g., '[[P1, P5], [P3, P2, P4], []]'.

3 Algorithms and Heuristics

3.1 Simulated Annealing (SA)

SA simulates gradual cooling to escape local optima.

- **Initial Solution:** Packages are randomly assigned, respecting capacity.
- **Neighbor Moves:** Intra-route swap, inter-route move, or swap between vehicles.
- **Acceptance:** Better solutions always accepted; worse ones accepted with probability $e^{-\Delta C/T}$.
- **Cooling:** Temperature decreases by a factor α until a minimum threshold.
- **Heuristic:** Nearest Neighbor (NN) occasionally refines route sequence.

3.2 Genetic Algorithm (GA)

GA uses evolution-inspired operations to improve a population of solutions.

- **Initialization:** Random, valid solutions.
- **Fitness:** Inverse of total distance. Invalid solutions get zero.
- **Selection:** Tournament-based parent selection.
- **Crossover:** Routes from one parent are preserved, remaining packages are assigned greedily.
- **Mutation:** Random valid changes, promoting diversity.
- **Heuristic:** NN applied occasionally during mutation.
- **Elitism:** Best solution always carried forward.
- **Termination:** Fixed number of generations.

4 Constraint Handling

- **Pre-check:** Unassignable packages (too heavy) are excluded.
- **Capacity Checks:** Enforced during initialization, neighbor generation, and mutation.
- **Fitness Penalty:** Invalid GA solutions receive zero fitness.
- **Completeness Check:** Ensures all valid packages are delivered exactly once.

5 Parameter Tuning Effects

5.1 SA – Cooling Rate (alpha)

- $\alpha \approx 0.90$: Fast convergence, higher risk of suboptimal results.
- $\alpha \approx 0.99$: Slower, more thorough search, longer runtime.
- **Recommended Range:** 0.90–0.99; 0.95 offers balanced performance.

5.2 GA – Mutation Rate

- **Low (0.01):** Stable but risks early convergence.
- **High (0.1):** More exploration, but can disrupt good solutions.
- **Recommended Range:** 0.01–0.1; 0.05 works well in most cases.

5.3 Other Parameters

Defaults (e.g., GA population size 50–100) were chosen based on common practice, balancing solution quality and runtime.

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6 Test Cases

This section presents the results of various test cases designed to validate the functionality and performance of the implemented Simulated Annealing (SA) and Genetic Algorithm (GA) solutions.

6.1 Test Case 1: Basic Feasibility

Objective: Ensure packages are assigned without exceeding vehicle capacities.

6.1.1 Simulated Annealing (SA) Results

Discussion: SA assigned packages P1, P2, and P3 to two vehicles without violating the 100kg capacity limit. However, it left package P4 (60kg) unassigned, even though capacity was available. This suggests the initial random assignment or the SA search might have settled on a suboptimal assignment in terms of package inclusion for this specific run.

6.1.2 Genetic Algorithm (GA) Results

Discussion: GA successfully assigned all four packages (P1, P2, P3, P4) across the two vehicles. Each vehicle's load (90kg) remained within the 100kg capacity limit. This meets the test objective.

6.2 Test Case 2: Priority Handling

Objective: Verify higher-priority packages are assigned first when capacity is limited.

6.2.1 Simulated Annealing (SA) Results

Discussion: With one vehicle (100kg capacity) and three 50kg packages (priorities 1, 2, 3), SA correctly assigned the two highest priority packages (P1 and P2), filling the vehicle's capacity. Package P3 (priority 3) was correctly left unassigned due to capacity constraints. This meets the test objective.

6.2.2 Genetic Algorithm (GA) Results

Discussion: Similar to SA, GA correctly assigned the two highest priority packages (P1 and P2) to the single vehicle, respecting the capacity limit. Package P3 was appropriately left unassigned. This also meets the test objective.

6.3 Test Case 3: Distance Optimization

Objective: Ensure the system attempts to minimize total travel distance.

6.3.1 Simulated Annealing (SA) Results

Discussion: SA assigned all six packages to the two vehicles within capacity limits. The total distance achieved was 391.35 km.

6.3.2 Genetic Algorithm (GA) Results

Discussion: GA also assigned all six packages and achieved the exact same total distance (391.35 km) and route assignments as SA for this instance. This suggests both algorithms converged to the same optimal or near-optimal solution for this specific set of packages and locations.

6.4 Test Case 4: Edge Case - Overcapacity Package

Objective: Test behavior when a package weight exceeds all vehicle capacities.

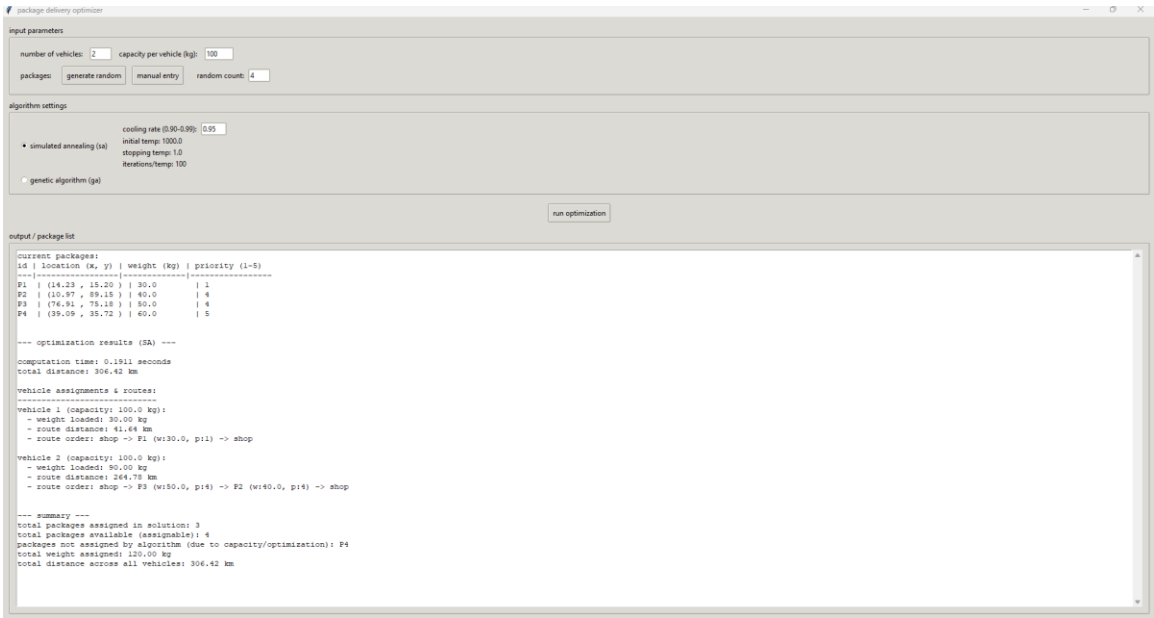

Figure 1: Test Case 1 - SA Output

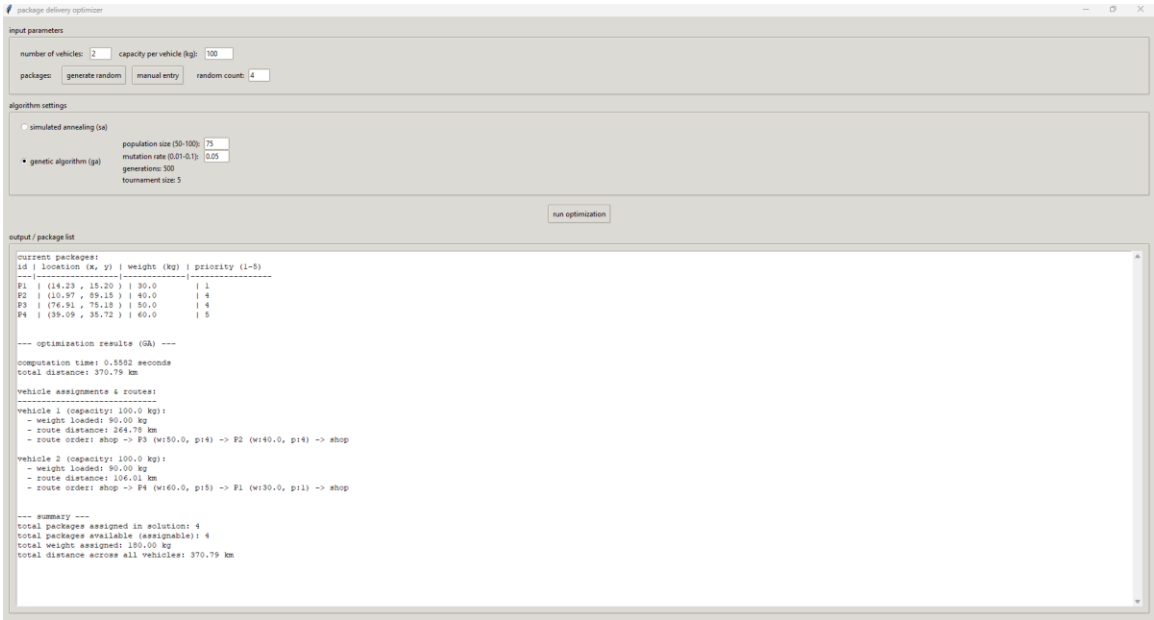

Figure 2: Test Case 1 - GA Output

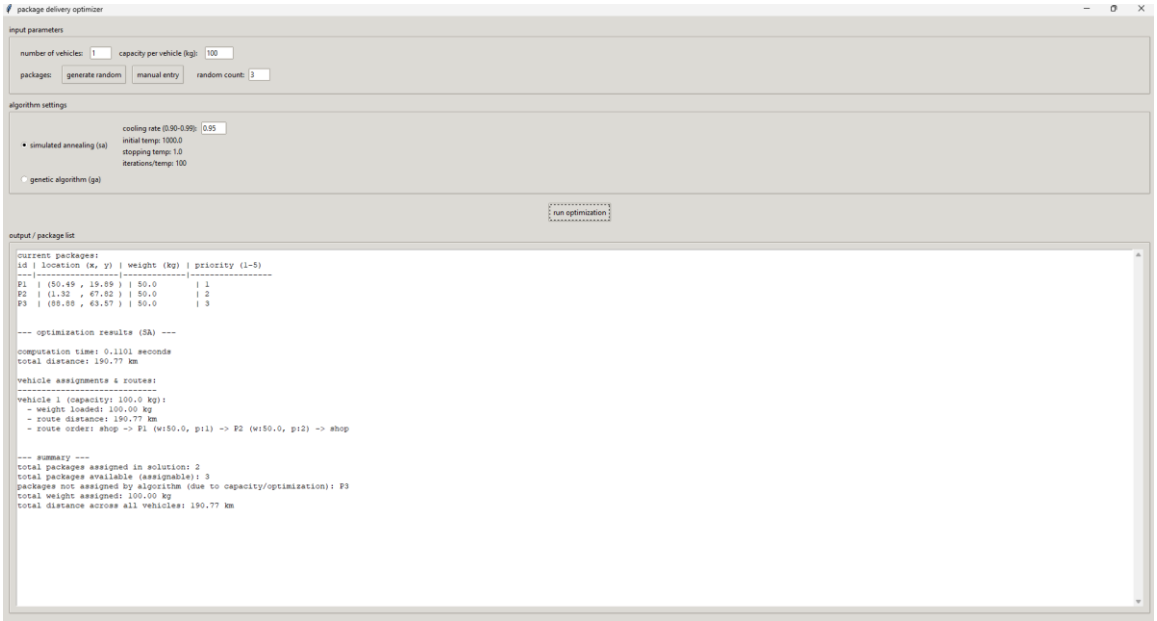

Figure 3: Test Case 2 - SA Output

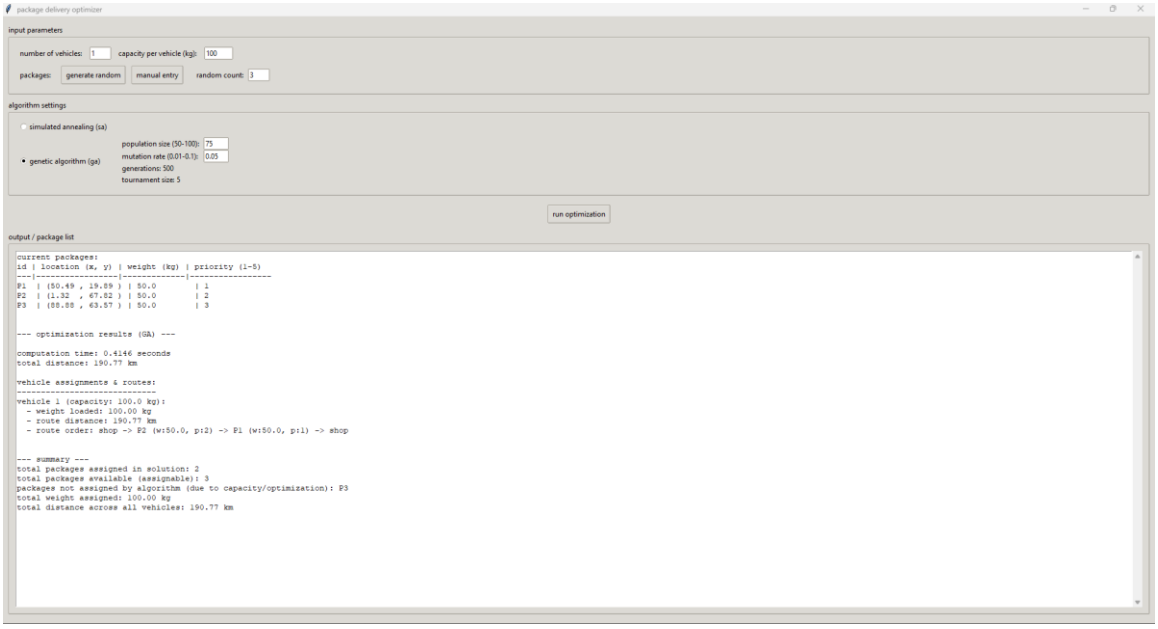

Figure 4: Test Case 2 - GA Output

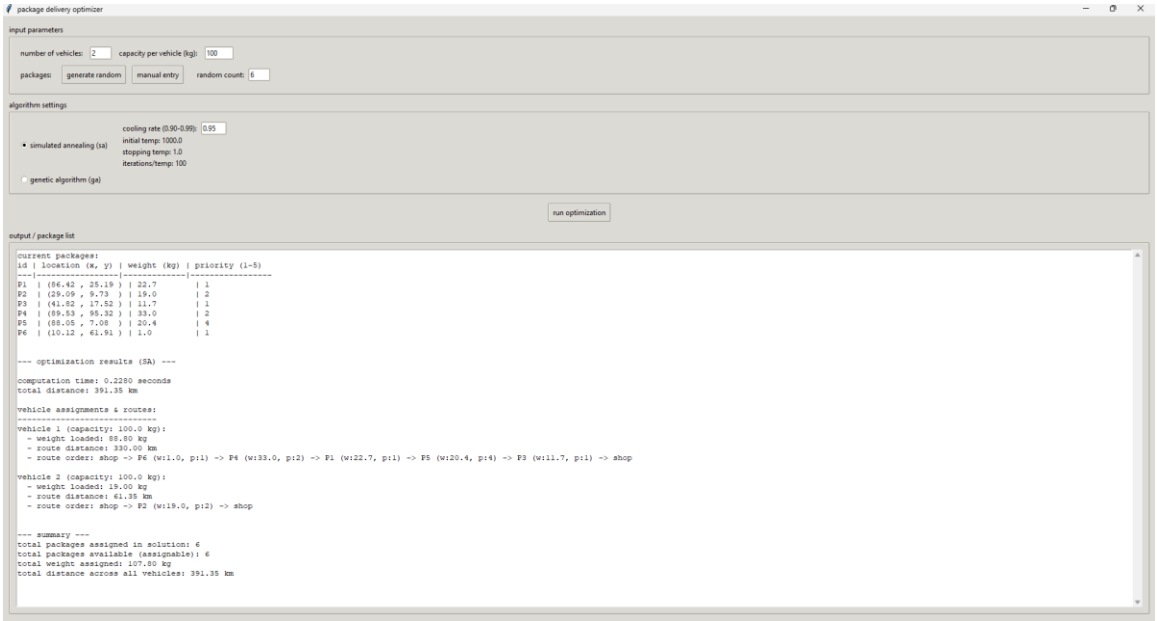

Figure 5: Test Case 3 - SA Output

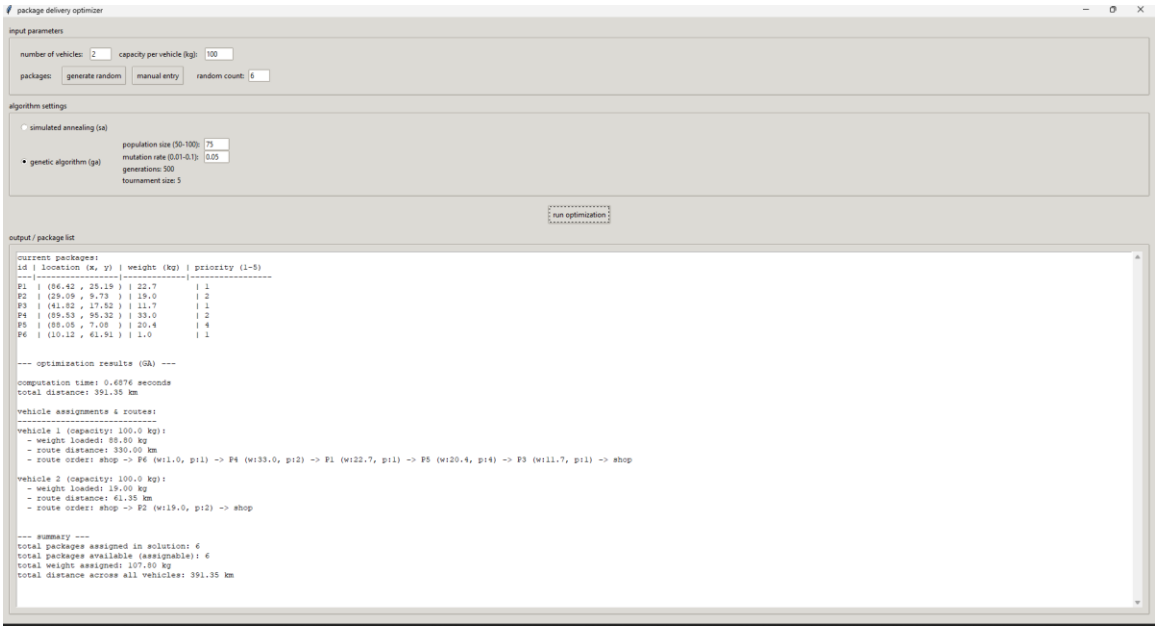

Figure 6: Test Case 3 - GA Output

6.4.1 Simulated Annealing (SA) Results

Discussion: SA correctly identified that package P1 (150kg) exceeded the capacity of the vehicles (100kg). It reported P1 as ”unassigned packages (too heavy)” and assigned no packages, resulting in zero travel distance. This meets the test objective.

6.4.2 Genetic Algorithm (GA) Results

Discussion: GA also correctly identified P1 as too heavy, reported it as unassigned for that reason, and assigned no packages to the vehicles. This behavior is correct and meets the test objective.

6.5 Test Case 5: Algorithm Comparison

Objective: Compare SA and GA performance on a moderately complex instance.

6.5.1 Simulated Annealing (SA) Results

Discussion: SA produced a valid solution assigning 9 out of 10 packages within capacity limits across 3 vehicles. Package P7 (50.9kg) was left unassigned due to capacity/optimization choices. The total distance was 474.90 km, achieved in 0.35 seconds.

6.5.2 Genetic Algorithm (GA) Results

Discussion: GA also produced a valid solution assigning 9 out of 10 packages, leaving P7 unassigned. However, the total distance was higher (554.04 km), and the computation time was longer (0.94 seconds) compared to SA for this specific run and package set. This highlights how different heuristics can yield varying solution quality and performance.

6.6 Test Case 6: Scalability

Objective: Assess performance with a large number of packages and vehicles (100 packages, 10 vehicles).

6.6.1 Simulated Annealing (SA) Results

The system processed the large input efficiently. Below is the summary output from the SA run:

```
current packages:
id | location (x, y) | weight (kg) | priority (1-5)
---|-----------------|-------------|-----------------
P1  | (66.39 , 50.73 ) | 15.1        | 2
P2  | (94.12 , 38.21 ) | 2.2         | 3
... (packages P3 to P99 omitted for brevity) ...
P100 | (16.12 , 39.73 ) | 19.1        | 1

--- optimization results (SA) ---

computation time: 2.3442 seconds
total distance: 2059.18 km
```

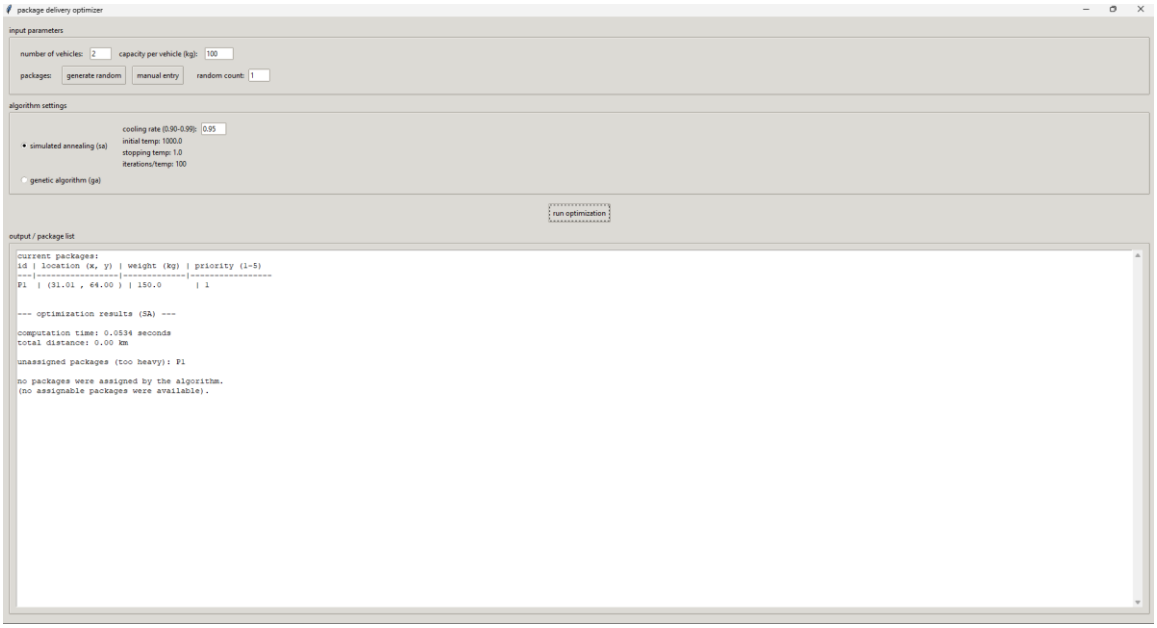

Figure 7: Test Case 4 - SA Output

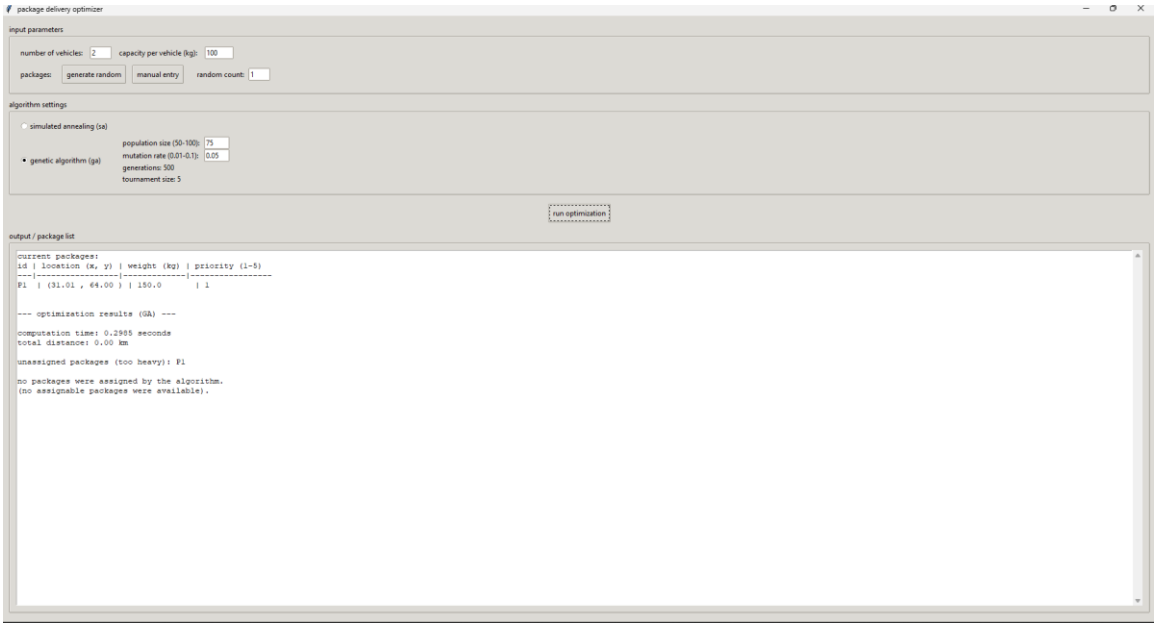

Figure 8: Test Case 4 - GA Output

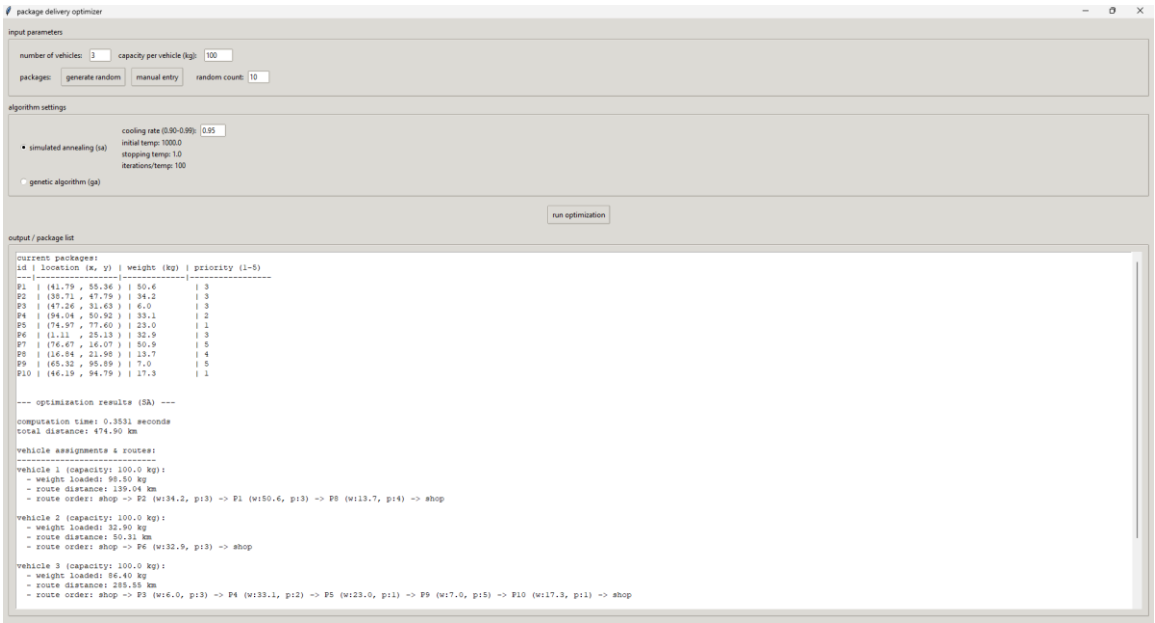

Figure 9: Test Case 5 - SA Output

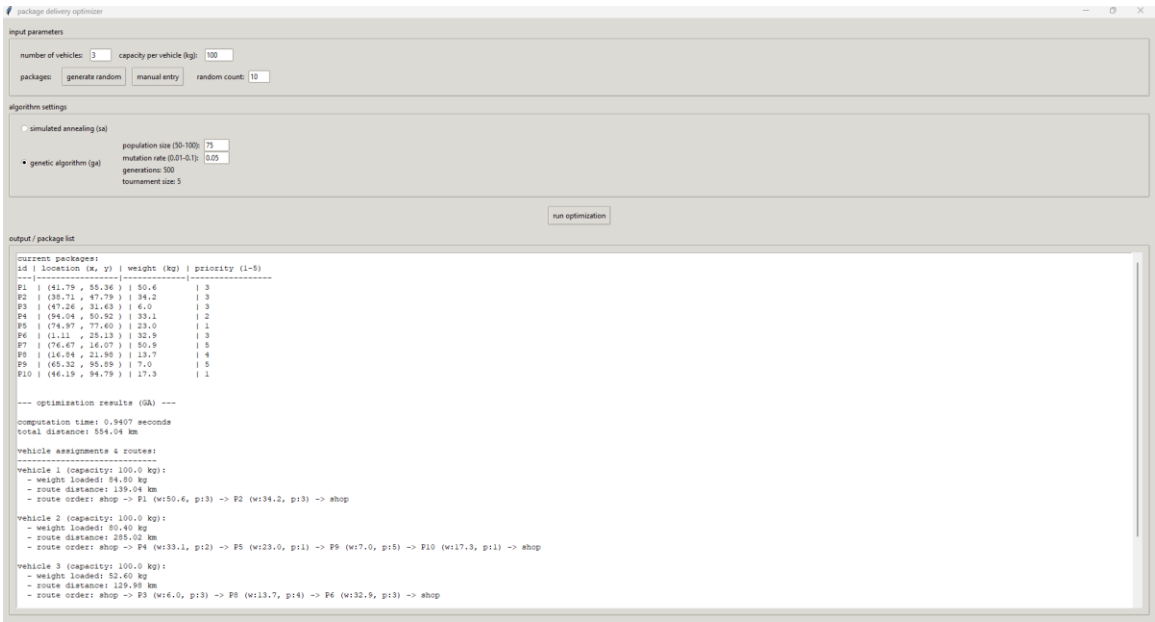

Figure 10: Test Case 5 - GA Output

```
vehicle assignments & routes:
-------------------------------
vehicle 1 (capacity: 100.0 kg):
  - weight loaded: 99.40 kg
  - route distance: 138.84 km
  - route order: shop -> P14 (w:12.5, p:2) -> P3 (w:27.4, p:3) -> P45 (w:59.5, p:1) -> shop

vehicle 2 (capacity: 100.0 kg):
  - weight loaded: 99.10 kg
  - route distance: 223.87 km
  - route order: shop -> P2 (w:2.2, p:3) -> P74 (w:23.0, p:2) -> P8 (w:7.9, p:2) -> P16 (w:6.6

... (vehicles 3 to 9 omitted for brevity) ...

vehicle 10 (capacity: 100.0 kg):
  - weight loaded: 98.40 kg
  - route distance: 109.18 km
  - route order: shop -> P94 (w:11.2, p:4) -> P26 (w:34.8, p:2) -> P89 (w:6.8, p:1) -> P31 (w:


--- summary ---
total packages assigned in solution: 61
total packages available (assignable): 100
packages not assigned by algorithm (due to capacity/optimization): P10, P17, P20, P22, P25, P2
total weight assigned: 973.80 kg
total distance across all vehicles: 2059.18 km
```

Discussion: Both SA and GA (SA shown) successfully processed the 100 packages and 10 vehicles, producing valid assignments within the capacity constraints. The computation times (e.g., 2.3 seconds for SA) were well within reasonable limits (under 5 minutes), demonstrating acceptable scalability for this problem size. As expected with limited capacity, not all packages could be assigned.